

LSMA-PQR: Lumbar Spine MRI Dataset

Multi-view Annotations · Pfirrmann Grading · Quantitative Measurements · NLP Reports

MULTI-VIEW IMAGING

- ▶ Axial: **3,090** (75%)
- ▶ Sagittal: **1,030** (25%)
- ▶ Levels: **D3 · D4 · D5**

$$\mathcal{I} = \mathcal{I}_{\text{ax}} \cup \mathcal{I}_{\text{sg}}$$

ANNOTATION FORMATS

- ▶ YOLO Polygons: **4,120**
- ▶ Semantic Masks: **4,120**
- ▶ Total Objects: **25,143**

$$\text{IoU} = 0.994 \pm 0.008$$

QUANTITATIVE

- ▶ IVD Heights: **1,545**
- ▶ Coordinates: **Pixel-level** (x, y)
- ▶ Error (MAE): **0.31 mm**

$$h_d = \|\mathbf{p}_{\text{sup}} - \mathbf{p}_{\text{inf}}\|_2$$

DUAL MODALITY

- ▶ T1-Weighted: **2,060**
- ▶ T2-Weighted: **2,060**
- ▶ Images per Patient: **8**

$$\forall p : |T1_p| = |T2_p| = 4$$

ANATOMICAL CLASSES

- ▶ Axial: **IVD · PE · SC**
- ▶ Sagittal: **VB · IVD · S · SC**
- ▶ Unique Structures: **7**

$$\mathcal{C}_{\text{ax}} \cap \mathcal{C}_{\text{sg}} = \emptyset$$

CLINICAL DATA

- ▶ Pfirrmann Grades: **1,545**
- ▶ Structured Reports: **515**
- ▶ Correlation: $\rho_s = 0.68$ ($p < .001$)

$$\mathcal{G} \in \{1, 2, 3, 4, 5\}$$

$\mathcal{N} = 515$
SYMPTOMATIC PATIENTS
 $|\mathcal{I}| = 4,120$
ANNOTATED IMAGES

✓ COMPLETE

◆ VERIFIED

★ 340+ FIXED

3 LEVELS

CC BY 4.0